3.5.2 The benefits of using modular programming include

- Less code to be written.
- A single procedure can be developed for reuse, eliminating the need to retype the code many times.
- Programs can be designed more easily because a small team deals with only a small part of the entire code.
- Modular programming allows many programmers to collaborate on the same application.
- The code is stored across multiple files.
- Code is short, simple and easy to understand.
- Errors can easily be identified, as they are localized to a subroutine or function.
- The same code can be used in many applications.
- The scoping of variables can easily be controlled.

3.5.3 Access Control

Access control is a security technique that regulates who or what can view or use resources in a computing environment. It is a fundamental concept in security that minimizes risk to the object. In other words access control is a selective restriction of access to data. IN Object oriented programming languages it is implemented through access modifiers. Classical object-oriented languages, such as C++ and Java, control the access to class members by public, private and protected keywords. Private members of a class are denied access from the outside the class. They can be handled only from within the class.

Public members (generally methods declared in a class) are accessible from outside the class. The object of the same class is required to invoke a public method. ***This arrangement of private instance variables and public methods ensures the principle of data encapsulation***.

Protected members of a class are accessible from within the class and are also available to its sub-classes. No other process is permitted access to it. This enables specific resources of the parent class to be inherited by the child class.

Python doesn't have any mechanism that effectively restricts access to any instance variable or method. Python prescribes a convention of prefixing the name of the variable or method with single or double underscore to emulate the behaviour of protected and private access specifiers.

All members in a Python class are public by default, whereas by default in C++ and java they are private. Any member can be accessed from outside the class environment in Python which is not possible in C++ and java.

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